

Hermes Agent Operations Manual

A Complete Guide to Operating the OAKAI Autonomous AI Platform

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01

Executive Brief

Purpose & Scope

This manual provides step-by-step instructions for operating the Hermes Agent platform — an autonomous AI agent orchestrator used to run scheduled tasks, generate reports, manage knowledge bases, and coordinate machine learning workflows across hybrid cloud and local environments.

Key Capabilities

Autonomous task scheduling via cron jobs

Multi-model routing (free-tier first: Nous Portal, OpenRouter, NVIDIA NIM)

Skill-based modular execution framework

Persistent memory + indexed knowledge bases

Approval-gated high-cost operations

Local stack fallback when cloud egress is blocked

Designed for solo operators, startup teams, and enterprise automation engineers who want reliable AI agents without vendor lock-in or runaway costs.

02

System Overview

Architecture & Components

The Hermes Agent platform consists of four core layers:

- **Terminal Layer** — Direct shell access for debugging, file ops, and ad-hoc commands
- **TUI Dashboard** — Interactive session management with live status monitoring
- **Cron Engine** — Scheduled autonomous jobs with model pinning and drift guard
- **Skill Framework** — Modular plugins for PDF generation, knowledge indexing, etc.

Current stack as of 2026-08-12:

Component	Provider	Model/Version
Interactive Model	Nous Portal	poolside/laguna-s-2.1:free
Cron Jobs	Nous Portal	tencent/hy3:free (drift-pinned)
Vision	NVIDIA	nemotron-nano-12b-v2-vl
PDF Engine	ReportLab	5.0.0 (via venv)
Knowledge Store	Markdown + PyPDF	Local /opt/data/knowledge/

03

Interface Tour

Navigating the Dashboard

Hermes exposes three primary interfaces:

03A

Terminal Commands

CLI control layer

- ``hermes cron list`` — View all scheduled jobs
- ``hermes cron run [id]`` — Execute a job immediately
- ``hermes cron edit [id] --model X`` — Repin a drifted cron job
- ``hermes status`` — Full environment health check
- ``hermes tools list`` — See enabled/disabled tool integrations

03B

TUI Dashboard

Interactive session view

Access via ``hermes`` (no subcommand). Shows:

- Active cron job queue (next runs, last execution)
- Gateway connection status (web tools, image gen, TTS/STT)
- Model/provider availability and rate limits
- Session-level memory usage and index health

03C

File System Layout

Where everything lives

Path	Purpose
/opt/data/knowledge/	Living knowledge base + logs
/opt/data/workspace/	Generated deliverables + INDEX.md
/opt/data/skills/PDF/	Canonical PDF design system
/opt/data/cron/jobs.json	Cron job configurations
/opt/data/memories/	Persistent memory + USER profile

04

Daily Operations

Routine Management Tasks

04A

Running Crons

Scheduling & Monitoring

- Check next scheduled runs via `hermes cron list`
- Verify model pins: all crons must show 'tencent/hy3:free'
- Inspect output files in knowledge/[stream]/[date].md
- Confirm INDEX.md auto-appends new entries daily
- Review RUN_LOG.md for error traces or skipped steps

04B

Updating INDEX.md

Keeping the map fresh

The workspace INDEX.md is the single source of truth for all deliverables. It auto-appends entries from each cron run. Manual review recommended weekly.

04C

Regenerating PDFs

Using canonical templates

- Always use oakai_report_generator.py (not ad-hoc scripts)
- Output goes to /opt/data/workspace/[name].pdf
- Footer: 'OAKAI Confidential | Page N | Internal Use Only'
- TOC must match actual page numbers after first build
- Tables wrapped in KeepTogether to prevent orphaned headers

05

Safety Protocols

Approval Gates & Cost Control

05A

Approval Gate

Preventing unauthorized actions

All crons operate in **read-only + local-write** mode. Any action requiring:

- External API calls beyond free-tier quotas
- Financial transactions or budget changes
- File modifications outside /opt/data/
- Spawning new subagents recursively

Must trigger interactive approval via `/approve` flag in chat session.

05B

Budget Guardrails

Free-tier enforcement

Action	Max Daily Budget	Approval Required
Cloud LLM calls	50/day (OpenRouter)	Yes (above free tier)
Local LLM inference	Unlimited	No
Image generation (FAL)	10/day	No (free tier)
Browser automation	30/day	No (free tier)
New cron job creation	\$0	Yes (manual review)

06

Troubleshooting

Common Errors + Fixes

06A

Model Drift / Cron Failure

Symptom: Cron shows 'Script Error' or runs without output
Cause: Model/provider drift — global config changed but cron wasn't repinned
Fix: Run ``hermes cron edit [id] --model tencent/hy3:free --provider nous``

06B

Egress Blocked

Symptom: Web search or API call fails silently
Cause: Cloudflare WAF or DNS blocking on target domain
Fix: Use local LLM (Qwen2.5-1.5B + bge-m3) as fallback; switch providers
Workaround: Route through Nous Portal gateway which handles WAF bypass

06C

Missing venv / Python Modules

Symptom: ImportError: No module named 'reportlab' or similar
Cause: Virtual environment not activated or reportlab not installed
Fix: Use ``~/tmp/reportlab-venv/bin/python`` explicitly for PDF generation
Permanent: Add to PATH or use system-wide install path

07

Appendices

Reference Materials

07A

Appendix A — Cron Job Inventory

Job ID	Name	Schedule	Output Path
2b39ab1514d2	learn-pensolar	15:00 + 22:00 MYT	pensolar/logs/
3dfaf435889a	mentor-ai-daily	07:00 / 11:00 / 15:00 / 19:00	mentor/daily_notes/
75e36f8dd14d	marketing-advisor-daily	Mon–Sat 06:00 MYT	marketing/
9dda7d6f3af5	workspace-cleanup-daily	10:00 MYT	Auto: cleanup-policy.conf
0ee860f4fb74	startup-catchup-enforcement	06:00 daily	Auto: re-pin + backfill

07B

Appendix B — File Path Map

All deliverables organized as follows:

/opt/data/knowledge/mentor/	AI concept lessons + student progress
/opt/data/knowledge/by_industry/solar_energy/pensolar/	PENSOLAR solar case study
/opt/data/knowledge/marketing/	Daily marketing briefs + LinkedIn guidance
/opt/data/knowledge/strategy/	Sunday COO strategy briefs
/opt/data/knowledge/templates/	Python generators for reports
/opt/data/workspace/	Generated PDFs + INDEX.md manifest